Does Twitter feel the war in real time?

VADER sentiment vs. Russia-Ukraine war events at CHTC scale

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The question

When a major Russia-Ukraine war event hits the wire, **does the sentiment of public tweets shift, and for how long?**

We score ~52 GB of tweets (70.9M tweets, 476 days) with VADER on the CHTC cluster, then run event-study regressions and change-point detection against a curated timeline of 21 major war events.

The data

- **Kaggle:** `bwandowando/ukraine-russian-crisis-twitter-dataset-1-2-m-rows`
- ~480 daily CSVs, packed into 39 multi-day chunks on CHTC
- **~52 GB total**, Feb 2022 to June 2023
- After scoring: **70,876,101 tweets across 476 days**
- Columns used: `text`, `tweetcreatedts`, `language`, `retweetcount`, `followers`

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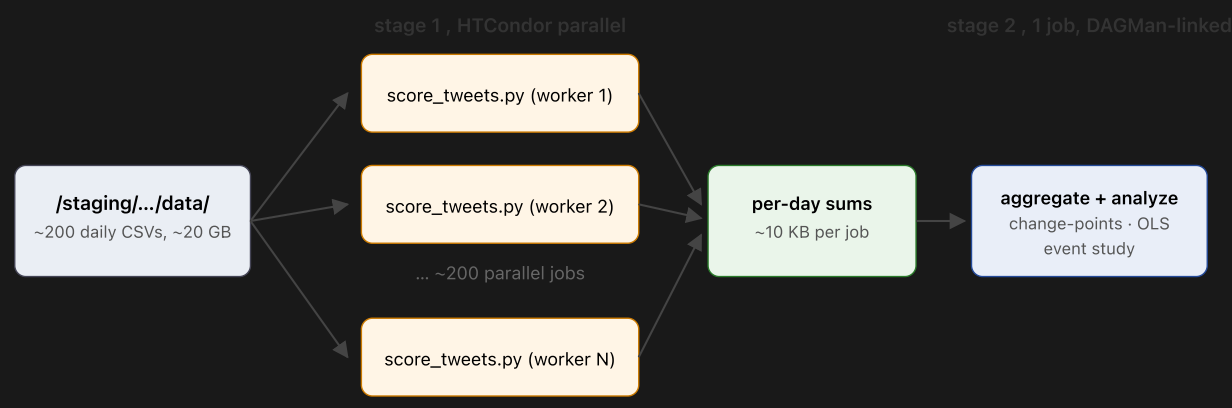
One line of code to read one file:

```
tweets = pd.read_csv("0819_UkraineCombinedTweetsDeduped.csv")
```

Parallel computation on CHTC



Sufficient statistics (sums, sums-of-squares) mean the aggregator never revisits raw tweets.

- **Stage 1**: 39 HTCondor jobs in parallel (one per chunk), Apptainer container from `/staging`
- **Stage 2**: single DAGMan-linked aggregator
- **Wall-clock**: ~1h40min end to end (calibration chunk: 7 min; largest chunks: ~50 min)
- Raw tweets never leave `/staging`; `learn` stays clean.

Method

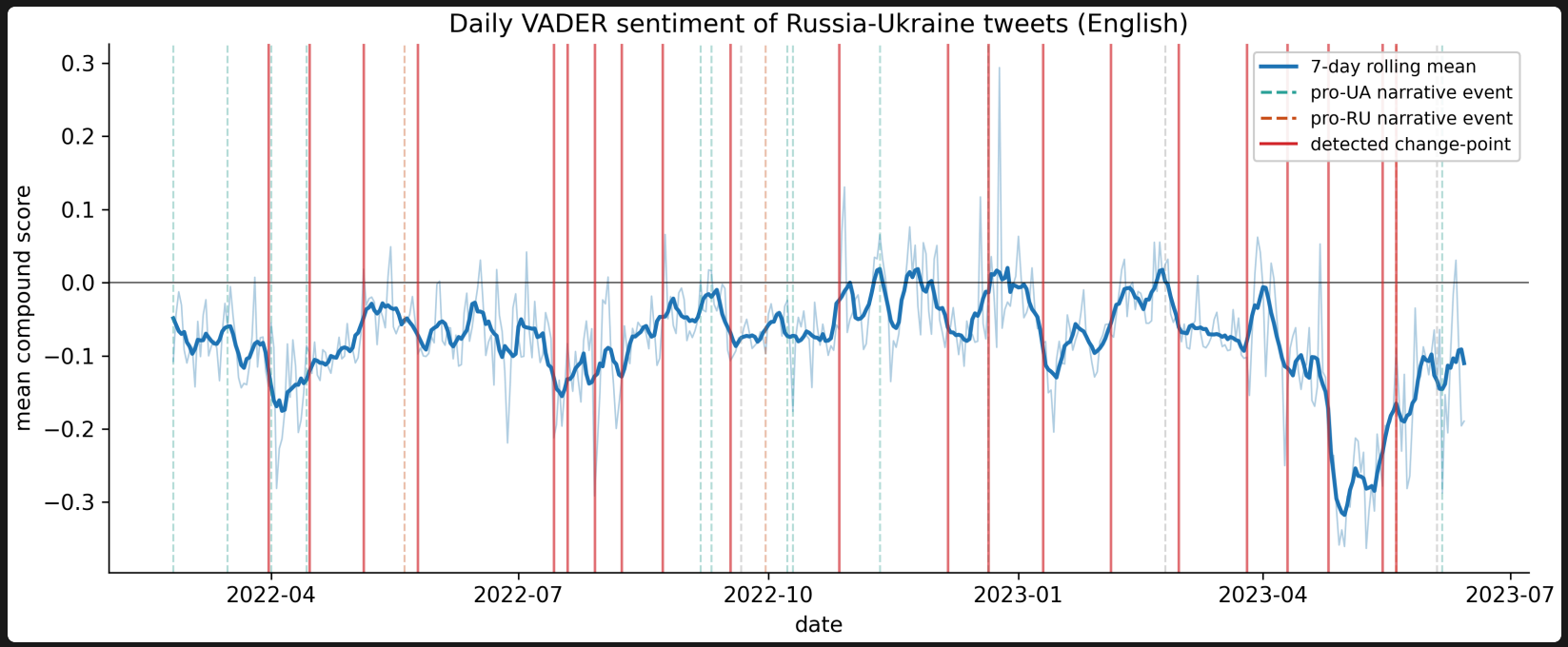
VADER compound score per tweet (-1 = very negative, +1 = very positive)

Each file produces per-(date, language) sums:

`sum(compound)` `sum(compound^2)`, `sum(followers*compound)` counts. These let the aggregator compute daily means, variances, and reach-weighting without re-reading the raw tweets.

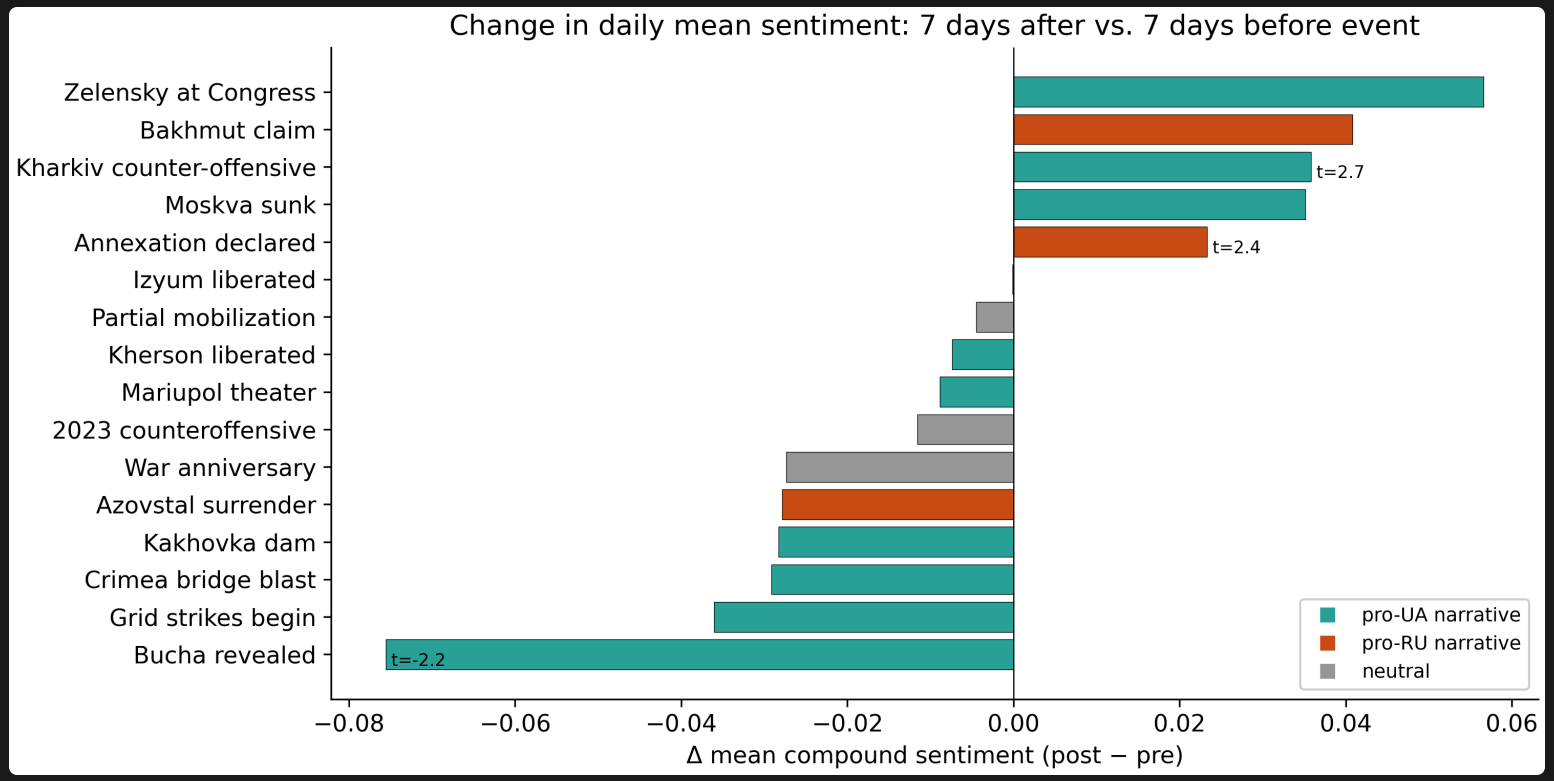
Three analyses on the daily time series:

Result 1: sentiment timeline



Blue = 7-day rolling mean (English); dashed = annotated events; red = PELT change-points. Daily mean drifts between -0.05 and -0.15; shorter dip at Bucha (Apr 2022), deepest excursion during Bakhmut period (spring 2023).

Result 2: event study



3 of 16 events pass |t| > 2: **Kharkiv counter-offensive** (+0.036, t=2.65), **Annexation declared** (+0.023, t=2.41), **Bucha revealed** (-0.076, t=-2.25).

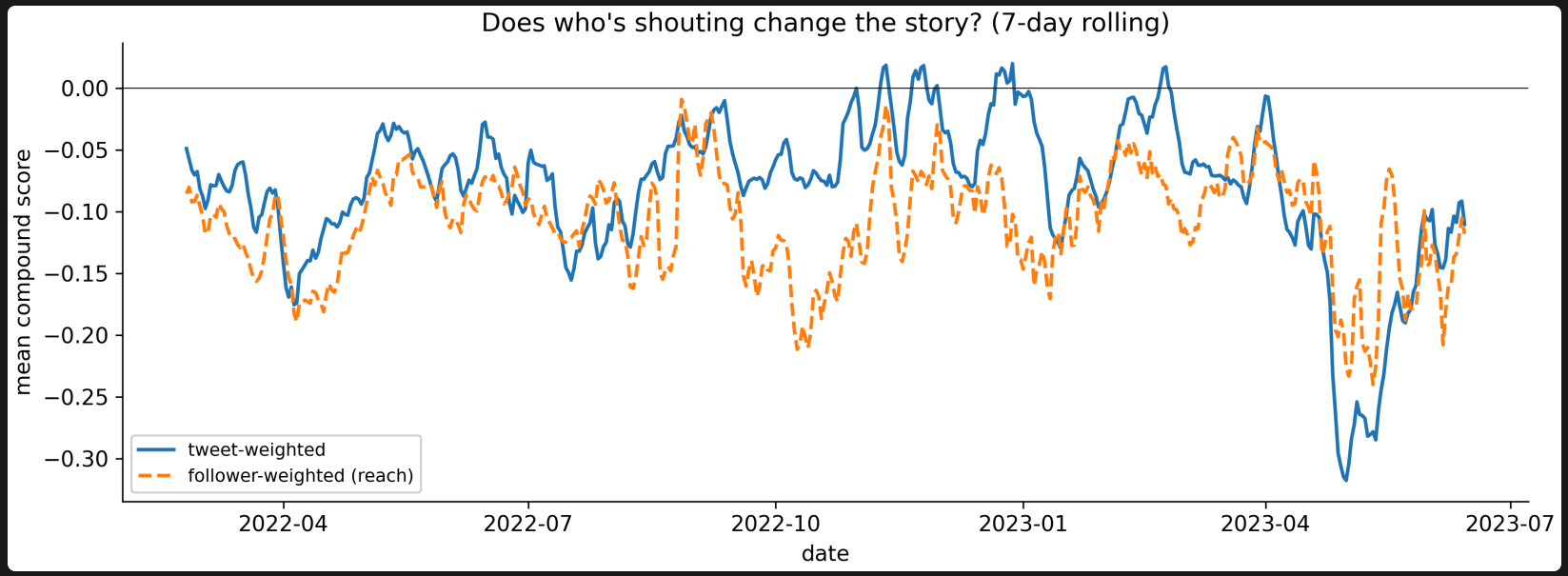
Result 3: regression + change-points

OLS (n=476): $R^2 = 0.092$, adj = 0.054, F = 2.44, $p = 0.0007$. Coefficients with $|t| > 2$:

Event	beta	p
Kherson liberated	**+0.119**	0.006
Kakhovka dam	**-0.108**	0.019
War anniversary	+0.092	0.036
Zelensky at Congress	+0.089	0.040

PELT finds 21 structural breaks; only 2022-12-21 (Zelensky at Congress) aligns with a labeled event.

Result 4: reach weighting



Follower-weighting shifts amplitude, not sign: loudest accounts mirror the median tweeter.

Takeaways + limitations

Found: events explain ~9% of daily variance. Kherson + Kakhovka register with the expected sign in the regression; Bucha is the clearest single-event dip. Most change-points don't match our event list.

Caveats: VADER is English-only (non-English looks muted); Durbin-Watson 0.70 means strong autocorrelation (OLS SEs understated); events cluster, so dummies are not clean causal effects; no bot filter.

Next: multilingual sentiment (XLM-R), bot filter, RDD around each event.

```
git clone https://github.com/matteso1/STAT405project.git
```